

CPS I — Exercise Sheet 4

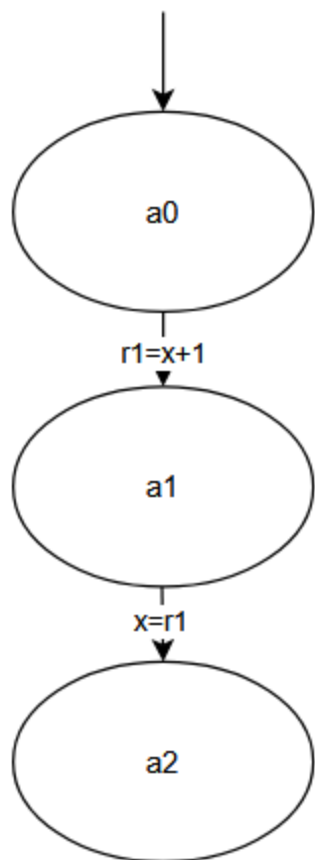
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Author: Zhanyu Dong, Xin Pu

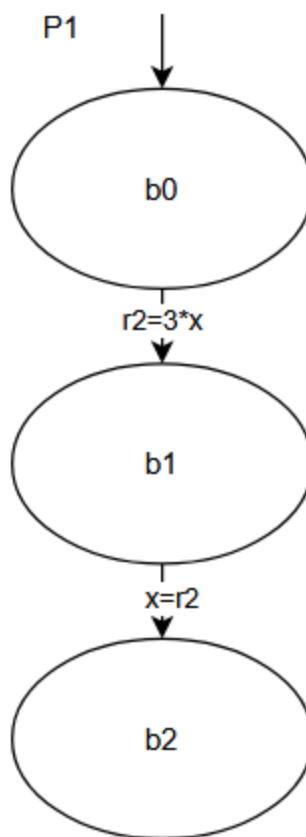
1	2	Σ
3.5/6	4.5/8	8/14

Exercise 1: Parallelism - Interleaving 3.5/6

a.



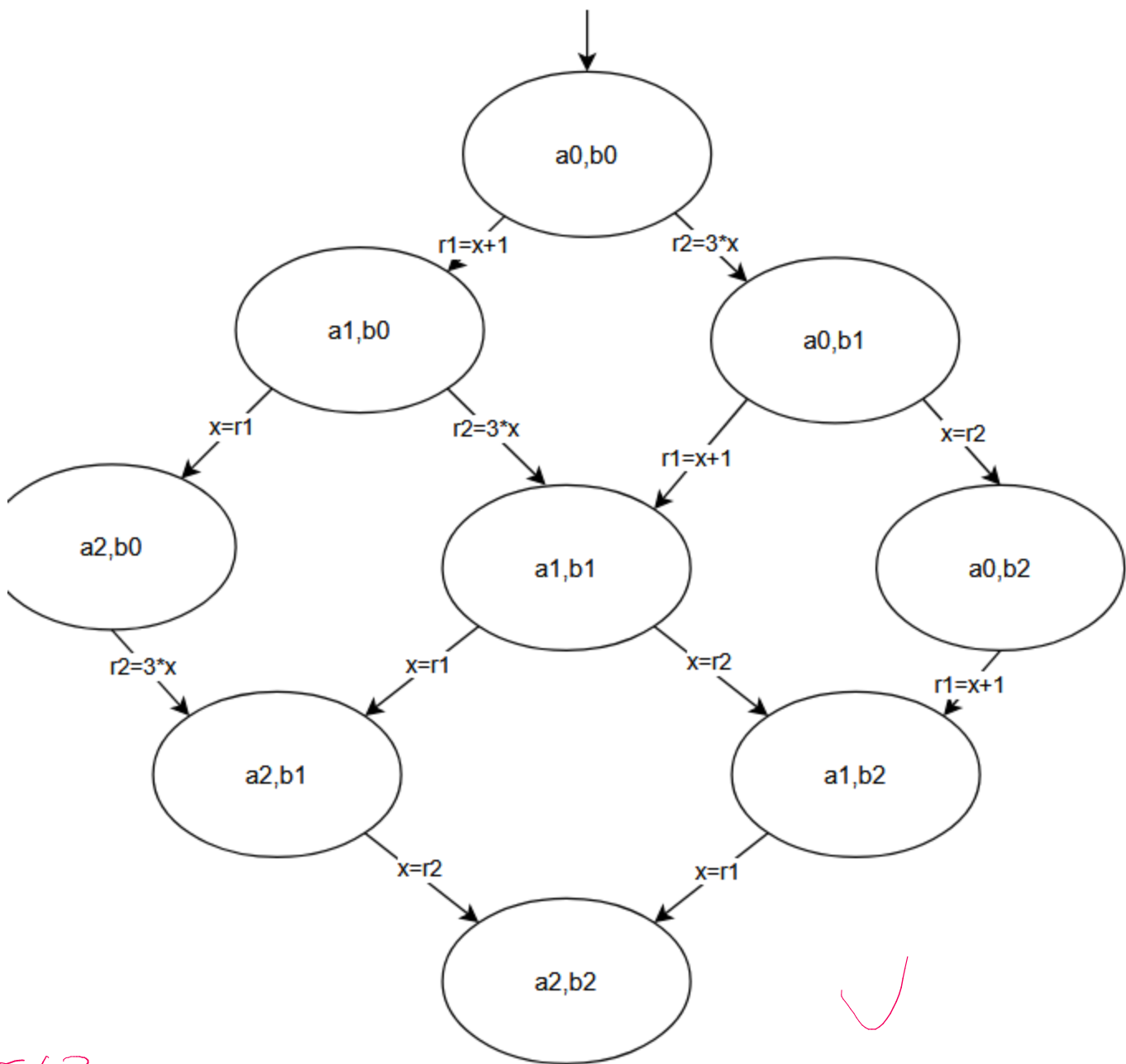
✓



✓

b.

2/2



0.5/3

c. The reachable state is (a2,b2), and the finally x value is (2,3,4,6)

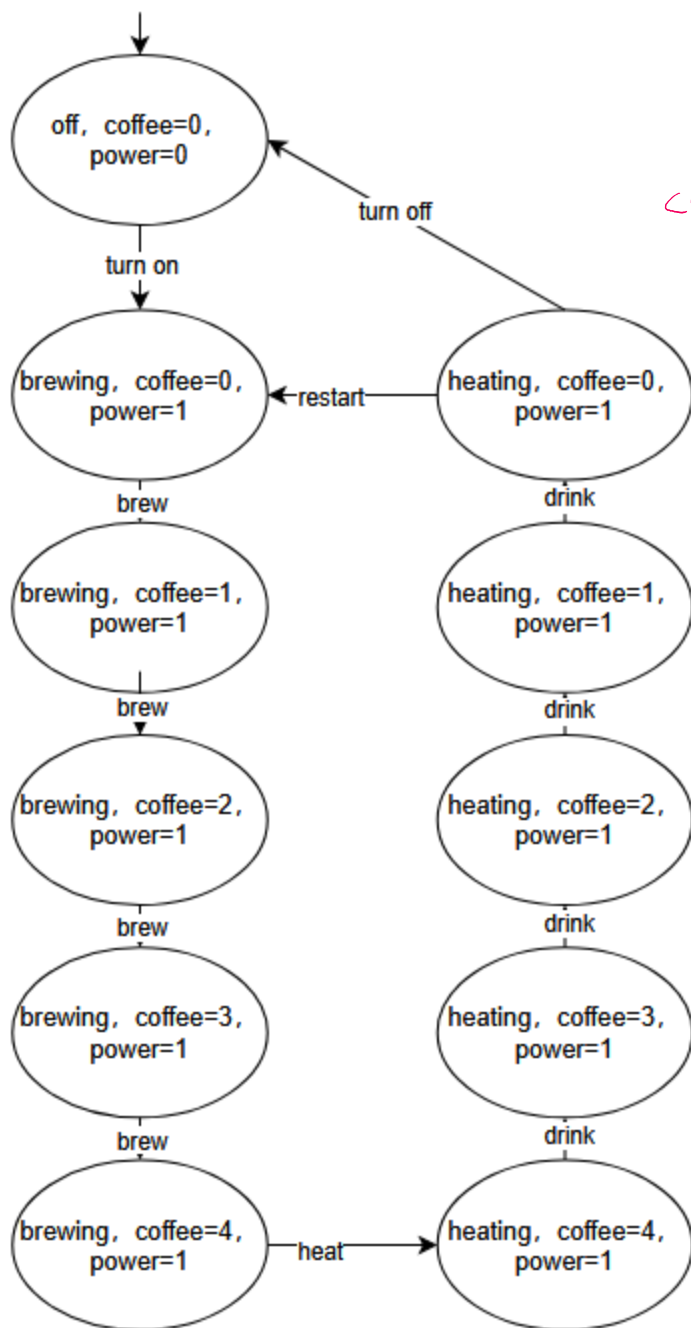
reachable part of $T_{P_1 \parallel P_2}$ missing $\downarrow$

I get what you mean,
but this looks like x is a 4-tuple
and has only one possible value,
namely $(2, 3, 4, 6)$

4.5/8

Exercise 2: Coffee Machine and Transition System

3.5/5



correct for (a),
labels missing for (b)

- $\langle \text{off}, \{0, 0\} \rangle \xrightarrow{\text{turn.on}} \langle \text{brewing}, \{0, 1\} \rangle$

- 0.5

$$\frac{s_1 \xrightarrow{\alpha} s'_1}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s'_1, s_2 \rangle}$$

$$s_1 = \langle \text{off}, 0, 0 \rangle$$

$$s'_1 = \langle \text{brewing}, 0, 1 \rangle$$

SAS - rules for TS(PH):

$$\frac{\ell \xrightarrow{g:\alpha} \ell' \wedge \eta \models g}{\langle \ell, \eta \rangle \xrightarrow{\alpha} \langle \ell', \text{Effect}(\alpha, \eta) \rangle}$$

$$\alpha = \text{turn.on}$$

- $\langle \text{brewing}, \{0, 1\} \rangle \xrightarrow{\text{brew}} \langle \text{brewing}, \{1, 1\} \rangle$

- 0.5

$$\frac{s_1 \xrightarrow{\alpha} s'_1}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s'_1, s_2 \rangle}$$

$$s_1 = \langle \text{brewing}, \text{coffee} = 0, \text{power} = 1 \rangle$$

$$s'_1 = \langle \text{brewing}, \text{coffee} = 1, \text{power} = 1 \rangle$$

$$\alpha = \text{brew}$$

- $\langle \text{brewing}, \{4, 1\} \rangle \xrightarrow{\text{heat}} \langle \text{heating}, \{4, 1\} \rangle$

- 0.5

$$\frac{s_1 \xrightarrow{\alpha} s'_1}{\langle s_1, s_2 \rangle \xrightarrow{\alpha} \langle s'_1, s_2 \rangle}$$

$$s_1 = \langle \text{brewing}, \text{coffee} = 4, \text{power} = 1 \rangle$$

$$s'_1 = \langle \text{heating}, \text{coffee} = 4, \text{power} = 1 \rangle$$

$$\alpha = \text{heat}$$

Reasons the Given Transitions Are Invalid

- $\langle \text{off}, \{0, 0\} \rangle \xrightarrow{\text{heat}} \langle \text{heating}, \{0, 0\} \rangle$:

The "heat" action is only associated with the "brewing" location (no "heat" transition exists from the "off" location in the program graph). SOS rules require actions to match the current location,

so this transition is invalid. What do you mean by "match the current location"?

The SOS rule requires that: $\text{off} \xrightarrow{\text{heat}} \text{heating} \wedge P \vdash 0, C \vdash 0 \models \text{?}$ for the transition to exist

- $\langle \text{brewing}, \{4, 1\} \rangle \xrightarrow{\text{brew}} \langle \text{brewing}, \{5, 1\} \rangle$:

The guard condition for "brew" is $\text{coffee} < 4$; here, the coffee amount is 4, so the guard condition fails. SOS rules require guard conditions to hold for transitions to fire, so this transition is invalid.

What do you mean by "transitions to fire"?

Part (b)

7/3

We label the transition system with atomic propositions (e.g., $P_{\text{off}} : \text{power} = 0$, $P_{\text{coffee}0} : \text{coffee} = 0$) and verify the properties below: *I don't see labels in your TS. -0.5*

(i)

True:

Only the "off" location has $\text{power} = 0$, and the coffee amount at "off" is always 0 (the initial state and the guard condition for the "turn.off" transition require $\text{coffee} = 0$). ✓

(ii)

False:

What about $\langle \text{heating}, \text{coffee} = 2, \text{power} = 1 \rangle$?

When $\text{coffee} = 2$, the current location is "brewing" (since $2 < 4$). The only enabled action is "brew", which increases the coffee amount to 3 ($2 + 1$). It cannot jump directly to 4, so the next step can only be 3. *ok but then the property would be true? insufficient explanation: -0.5*

↳ "brewing" is not a location. Did you mean the state $\langle \text{brewing}, \text{coffee} = 2, \text{power} = 1 \rangle$? there are multiple states that contain "brewing".

(iii)

True: *there are no guards in TS -0.5*

The guard condition for "brew" is $\text{coffee} < 4$, so "brew" stops firing once $\text{coffee} = 4$. All other actions (e.g., "drink") either decrease or preserve the coffee amount, so $\text{coffee} \leq 4$ at all times.

(iv)

False:

path: state sequence: $s_0 s_1 s_2 \dots$

execution: alternating sequence of states and actions: $s_0 \xrightarrow{\alpha_1} s_1 \xrightarrow{\alpha_2} s_2 \xrightarrow{\alpha_3} \dots$

what is an execution path?

"off", "heating" etc are not states of the TS. States in the TS include evaluations.

An execution path (e.g., "off $\rightarrow$ brewing $\rightarrow$ brew $\times 4 \rightarrow$ heating $\rightarrow$ drink $\times 4 \rightarrow$ restart $\rightarrow$ brewing $\rightarrow \dots$ ") is only off initially and stays powered on afterward. Thus, $\text{power} = 0$ does not occur infinitely often.

(v)

True:

do you mean "no self-loop in $\langle \text{off}, c=0, p=0 \rangle$ "?

When $\text{coffee} = 0$, execution paths must progress (the system cannot remain indefinitely in the initial "off" state without triggering actions). For example: (1) off $\rightarrow$ turn.on (step 1: brewing, $\text{coffee} = 0$); (2) brewing $\rightarrow$ brew (step 2: brewing, $\text{coffee} = 1$). Even in longer valid paths, the coffee amount becomes positive within 3 steps. *I don't get this at all.*

insufficient explanation. -0.5

In general:

Please use more precise wording in your answers. Use terminology established in this course.